APEX INSTITUTE OF TECHNOLOGY (AIT-CSE)

Chandigarh University, Punjab — 140413

Finance Tracker Application

A Web-Based Platform for Finance Tracker Application

Full Stack Web Development

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PROBLEM STATEMENT

"In today's fast-paced economic environment, individuals often struggle to maintain a clear and accurate understanding of their personal financial health. Traditional methods of tracking income and expenditures, such as manual logging in notebooks or managing complex spreadsheets, are time-consuming, prone to error, and lack real-time insights. Without a centralized and intuitive system to categorize transactions and visualize spending habits, users find it difficult to identify unnecessary expenses, adhere to budgets, and achieve long-term financial goals. Therefore, there is a distinct need for a secure, accessible, and user-friendly web application that automates financial tracking, categorizes transactions, and provides visual analytics to empower users to make informed financial decisions."

Core Functionalities:

- **User Authentication**
  Users can securely register and log in to the system.
- **Add Transactions**
  Users can add income and expense details with relevant information.
- **View Transactions**
  Users can view a list of all their financial transactions.
- **Update/Delete Transactions**
  Users can edit or remove existing transactions.
- **Dashboard Overview**
  Displays a summary of income, expenses, and balance.
- **Protected Access**
  Only authenticated users can access their financial data.
- **Data Storage**
  All user and transaction data are securely stored in a database.

PROJECT DESCRIPTION

The **Personal Finance Tracker** is a full stack web application developed to simplify and improve the way individuals manage their personal finances. In today's digital era, many people face challenges in keeping track of their daily income and expenses due to the absence of an efficient and user-friendly system. This project addresses that issue by providing a centralized platform where users can monitor and control their financial activities with ease.

The application allows users to create an account and securely log in using authentication mechanisms. This ensures that each user's financial data remains private and protected. After successful login, users are redirected to a personalized dashboard where they can manage their financial records.

One of the key features of the system is the ability to add transactions, where users can record both income and expenses along with relevant details such as amount, type, and description. The application also provides options to view all transactions in an organized manner, making it easier for users to analyze their spending patterns. Additionally, users can update or delete transactions whenever required, ensuring flexibility and accuracy of data.

The dashboard provides a summarized view of financial data, including total income, total expenses, and the overall balance. This helps users gain better insights into their financial status and supports informed decision-making. The user interface is designed to be simple, responsive, and easy to navigate, enhancing the overall user experience.

From a technical perspective, the frontend of the application is built using React, which enables dynamic and interactive user interfaces. The backend is developed using Node.js and Express, which handle server-side operations, API requests, and business logic. The application uses a relational database managed through an ORM to store user details and transaction records efficiently.

Overall, the Personal Finance Tracker serves as a practical solution for individuals who want to maintain financial discipline, track their expenses, and improve their financial planning through a secure and reliable system.

HARDWARE / SOFTWARE REQUIREMENTS

Hardware Requirements

Component	Minimum Specification	Recommended Specification
Processor (CPU)	Intel Core i3 (2.0 GHz or higher)	Intel Core i5/i7 (3.0 GHz or higher)
RAM	4 GB	8 GB or more
Hard Disk Storage	256 GB HDD	512 GB SSD
Display	1366 × 768 resolution	1920 × 1080 Full HD
Keyboard & Mouse	Standard Input Devices	Standard Input Devices
Network	Basic Internet Connection (10/100 Mbps)	High-Speed Broadband Connection
Power Backup	Optional	UPS recommended

Software Requirements

Category	Specification

Operating System	Windows 10 / Windows 11 / Ubuntu 22.04 LTS
Frontend Framework	React.js (Version 18 or above)
Backend Framework	Node.js with Express.js (Node v18 or above)
Database	MySQL / PostgreSQL
Authentication Library	JSON Web Token (JWT)
ORM Tool	Sequelize
Authentication Library	JSON Web Token (JWT)
State Management	Context API (or Redux Toolkit if used)
Development IDE	Visual Studio Code
API Testing Tool	Postman
Version Control System	Git & GitHub
Package Manager	npm (Node Package Manager)
Browser	Google Chrome / Mozilla Firefox
Deployment Platform	Vercel (Frontend), Render/Railway (Backend)
Documentation Tool	Microsoft Word / Google Docs

FRONT-END SCREENS

The front-end interface of the **Personal Finance Tracker** has been developed using React.js to provide a simple, responsive, and user-friendly experience. The interface is designed to allow users to easily manage their financial data, including adding, viewing, updating, and deleting transactions.

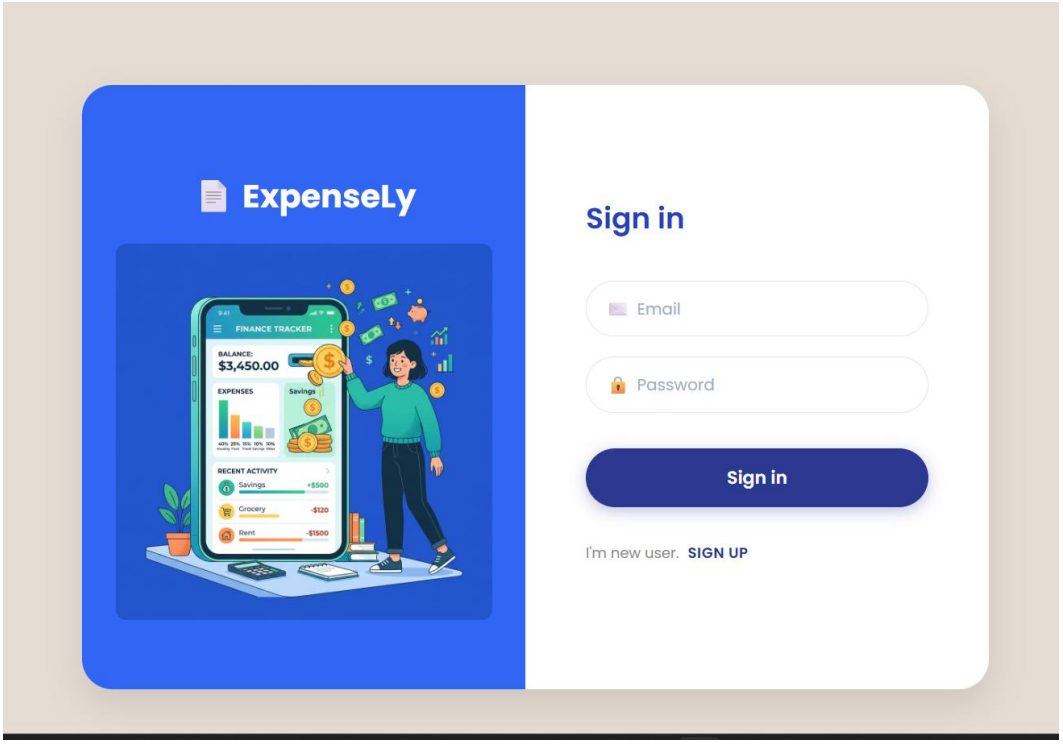
The application uses React Router DOM for smooth navigation between different pages such as login, registration, and dashboard. The user interface dynamically updates based on the authentication state, ensuring that only logged-in users can access the dashboard and financial features.

For state management, the application utilizes the Context API to handle user authentication and session data efficiently. API communication between the frontend and backend is managed using Axios, enabling seamless data exchange and real-time updates.

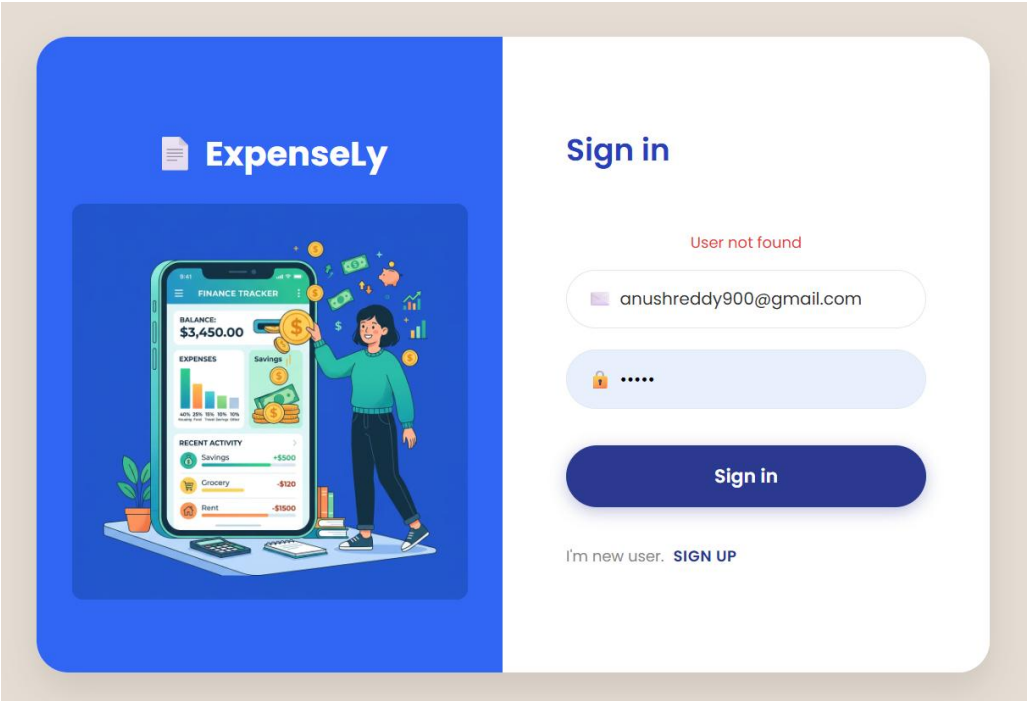
All UI components are designed with simplicity and usability in mind, ensuring that users can interact with the system without complexity. Additionally, protected routes are implemented to restrict unauthorized access, enhancing the security of the application.

Overall, the frontend provides an interactive and efficient platform for users to monitor and control their personal finances effectively.

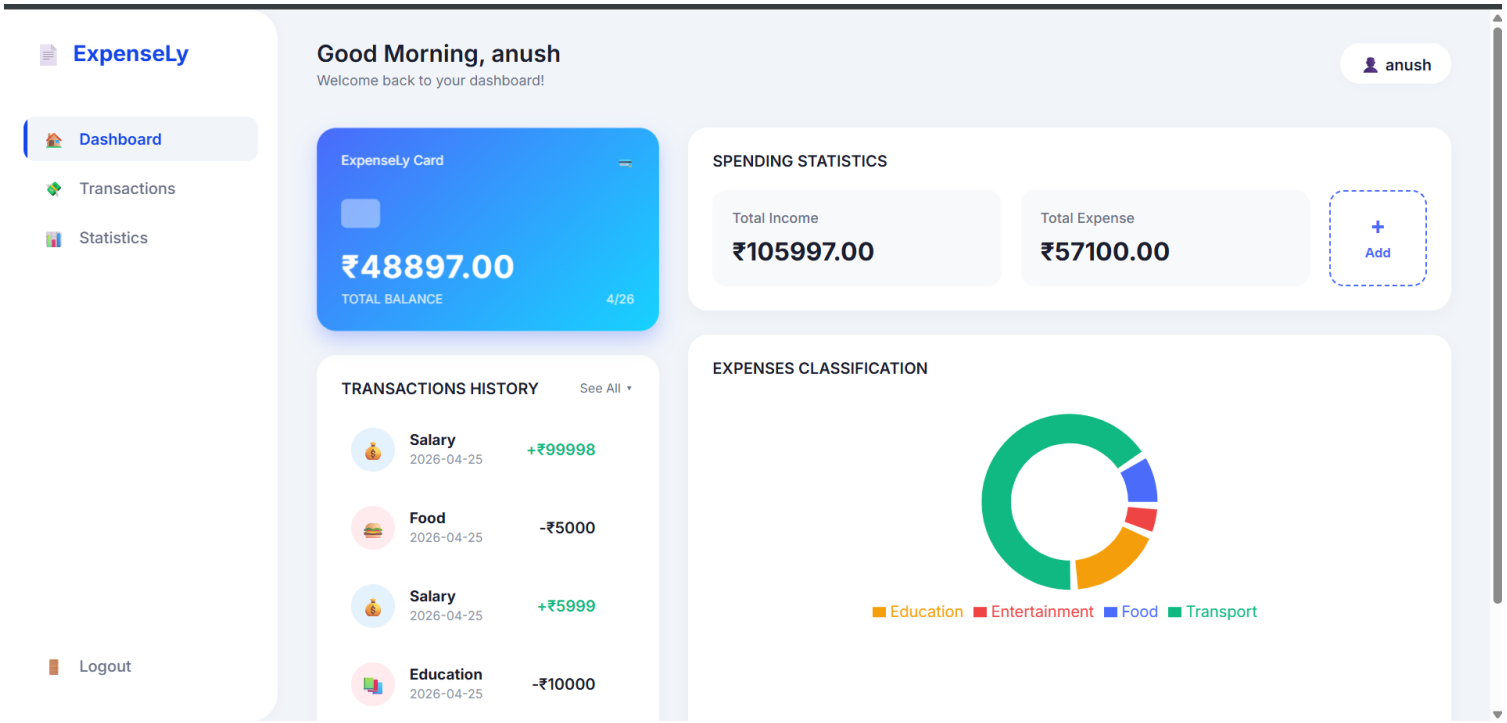
1. Login_Page

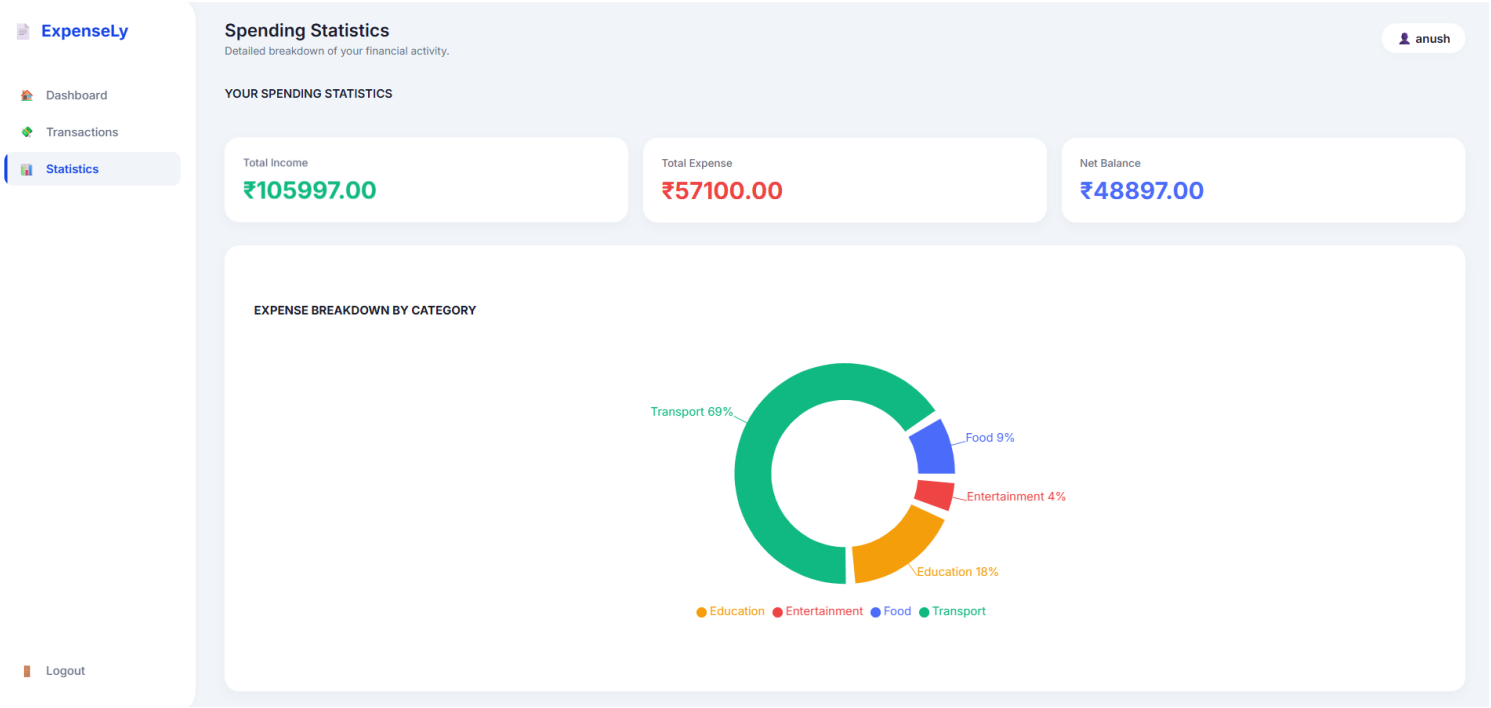


2. Access Denied Page

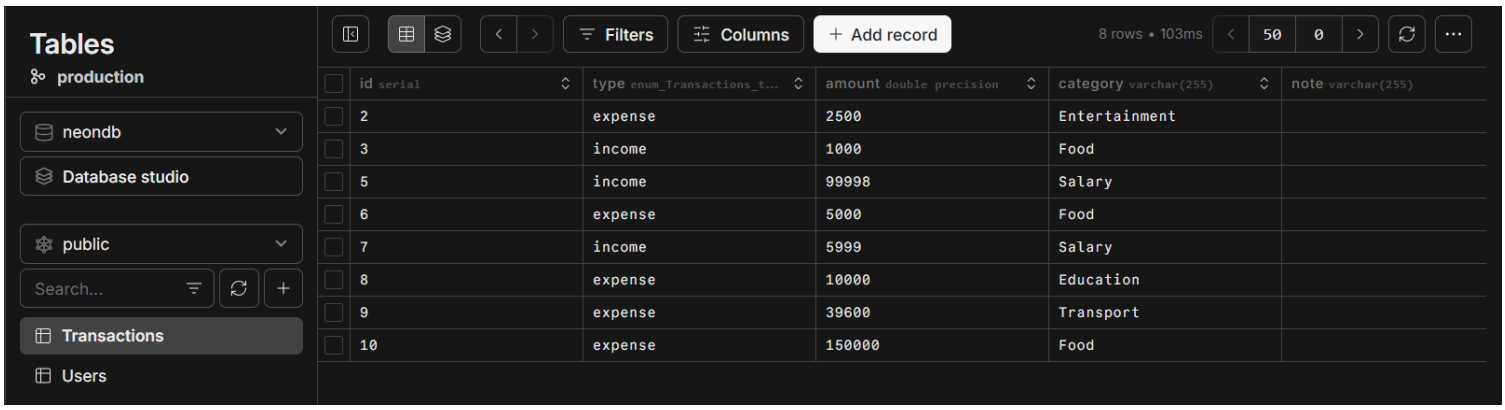


3. Admin Dashboard





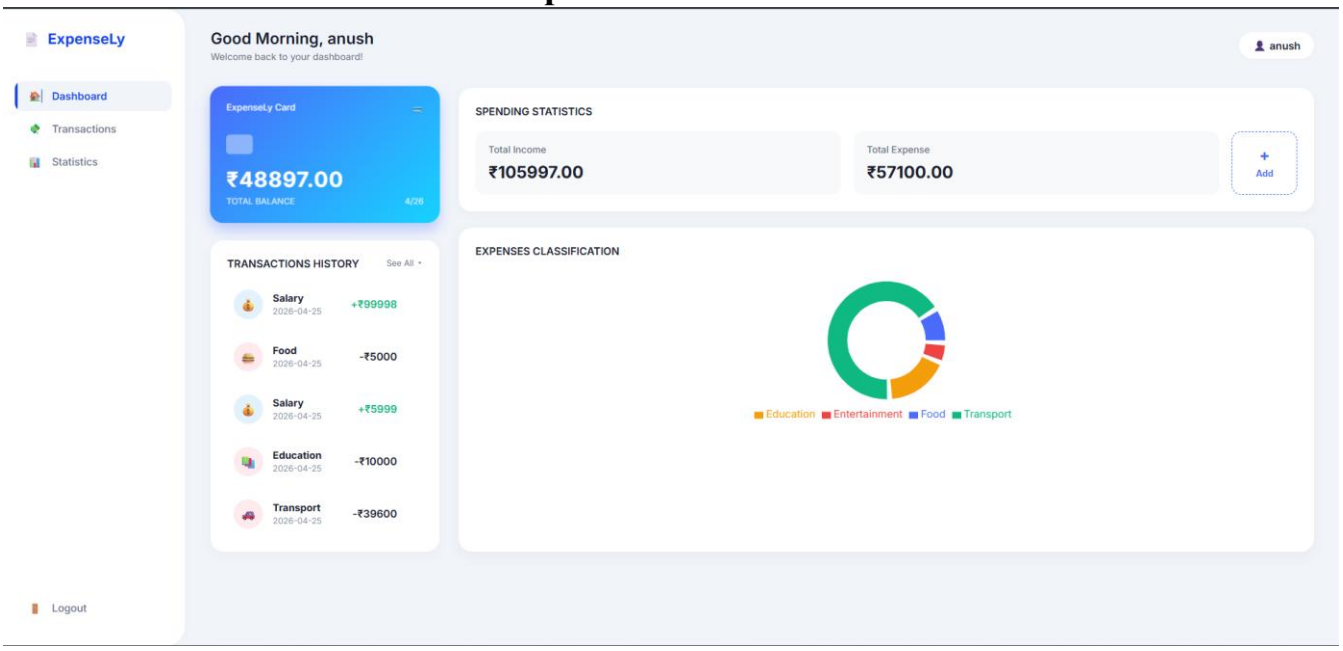
USER DATABASE



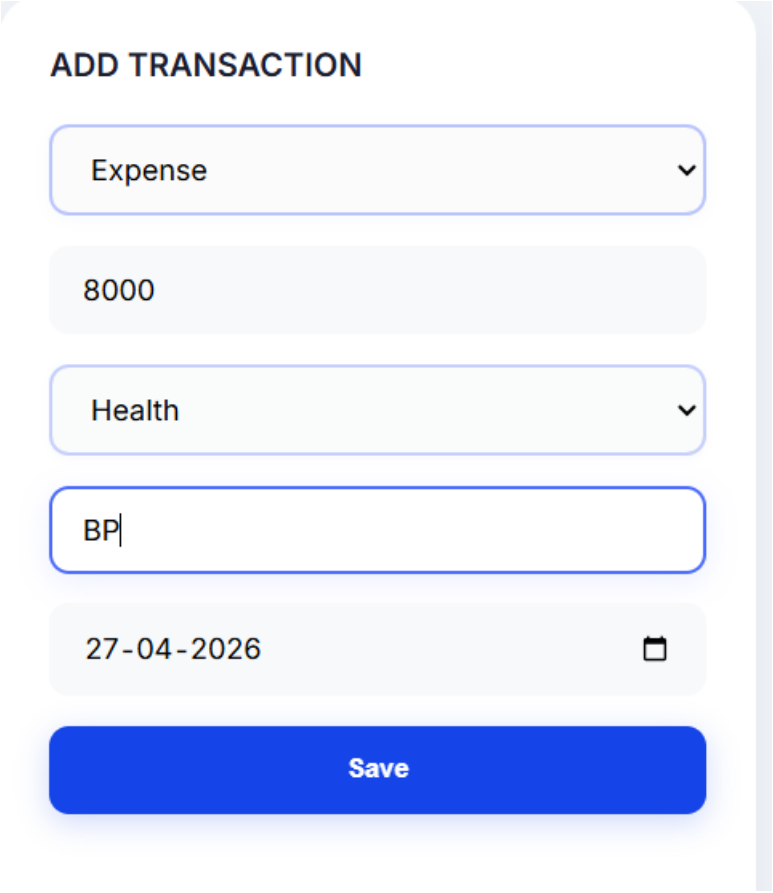
id serial	type enum_Transactions_t...	amount double precision	category varchar(255)	note varchar(255)
2	expense	2500	Entertainment	
3	income	1000	Food	
5	income	99998	Salary	
6	expense	5000	Food	
7	income	5999	Salary	
8	expense	10000	Education	
9	expense	39600	Transport	
10	expense	150000	Food	

OUTPUT SCREENS

1. Admin Dashboard Output



2. ADD Transaction



LIMITATION & FUTURE SCOPE

The **Personal Finance Tracker** project successfully demonstrates the implementation of a secure and user-friendly system for managing daily financial transactions using modern web technologies. It provides essential features such as user authentication and transaction management, ensuring efficient handling of personal financial data.

However, like any developing software system, there are certain limitations in terms of features, scalability, and advanced functionalities. These limitations also highlight potential areas for future improvements and enhancements to make the system more robust, scalable, and user-centric.

Limitations

- The application supports only basic financial tracking (income and expenses) without advanced analytics.
- It does not include data visualization features like charts or graphs for better insights.
- The system is dependent on an internet connection for accessing and managing data.
- Limited scalability for handling a very large number of users or transactions.
- Lacks features like budget planning, reminders, or financial goal tracking.

Future Scope

- Integration of data visualization tools (charts and graphs) for better financial analysis.
- Addition of budgeting and financial goal-setting features.
- Implementation of notifications and reminders for expenses or savings.
- Development of a mobile application for improved accessibility.
- Enhancement of security features such as multi-factor authentication.
- Integration with bank APIs for automatic transaction tracking.

GITHUB URL: https://github.com/harsh24jan/rbac

Benefits of the Project

- Easy Financial Management
  Helps users efficiently track their income and expenses in one place.
- User-Friendly Interface
  Simple and intuitive design makes it easy for anyone to use the system.
- Time-Saving
  Eliminates the need for manual record-keeping like notebooks or spreadsheets.
- Secure Data Handling
  User authentication ensures that financial data is protected and private.
- Real-Time Updates
  Transactions are updated instantly, providing accurate financial information.
- Better Financial Awareness
  Gives users a clear overview of their spending habits and balance.
- Accessibility
  Can be accessed anytime and anywhere through a web browser.

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